Algebra SEP Worksheet 4 7/28/22

Field and Galois Theory

1.

Give an example of an algebraic extension which has infinite degree.

Solution 1.

Consider the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$. It is by definition an algebraic extension. If it had finite degree d, then let ζ_p be a primitive pth root of unity with $p-1 > d$. Then $\mathbb{Q}(\zeta_p)$ has degree $p-1$ over $\mathbb{Q}$, so it can't be a subfield of $\overline{\mathbb{Q}}$, which is a contradiction. Thus, $\overline{\mathbb{Q}}/\mathbb{Q}$ has infinite degree.

$*$ **2.**

Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K:\mathbb{Q}] = 4$, and $\sqrt{-m} \in K$ for some positive integer m. Show that $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Solution 2.

This problem utilizes the same ideas as number 6 on the July 26 worksheet.

Since $K/\mathbb{Q}$ is Galois and degree 4, its Galois group is either $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. We can show that it is the latter by showing that K has two quadratic subfields, which means that $\mathrm{Gal}(K/\mathbb{Q})$ has two index 2 subgroups. We know that $\mathbb{Q}(\sqrt{-m}) \subseteq K$ is one quadratic subfield since $\sqrt{-m} \in K$ and the minimal polynomial of this element is $x^2 + m$. Another quadratic subfield is given by the fixed field of complex conjugation. Let $c : \mathbb{C} \to \mathbb{C}$ be complex conjugation. Then c is a nontrivial automorphism of K since c is not the identity on the subfield $\mathbb{Q}(\sqrt{-m})$. Since c has order 2, the fixed field of K by c is a quadratic subfield of K different from $\mathbb{Q}(\sqrt{-m})$.

$*$ **3.**

Let K be a field, and let $f(x) \in K[x]$ be an irreducible polynomial. Suppose that the splitting field L of $f(x)$ is a Galois extension of K (that is, f is separable). Let $\alpha \in L$ be one of the roots of f.

(a) Show that if $\mathrm{Gal}(L/K)$ is an abelian group, then $L = K(\alpha)$.

(b) Is the converse statement true? This is, suppose $L = K(\alpha)$; must $\mathrm{Gal}(L/K)$ be abelian?

Solution 3.

(a) Assume that $G = \mathrm{Gal}(L/K)$ is abelian. We know that $K(\alpha) \subseteq L$, so the Galois correspondence implies that there is a subgroup $H \leq G$ such that $H = \mathrm{Gal}(L/K(\alpha))$. Since G is abelian, H is a normal subgroup, so $K(\alpha)$ is a Galois extension of K. Since $f(x)$ is irreducible, it is the minimal polynomial of α over K, so f splits in $K(\alpha)$ since $K(\alpha)/K$ is Galois and contains a root of f so must contain all of the roots of f. Thus, $K(\alpha) = L$.

(b) This is not true. Let L be any finite Galois extension of $\mathbb{Q}$ with non-abelian Galois group. Then the primitive element theorem gives that $L = \mathbb{Q}(\alpha)$ for some $\alpha \in L$.

∗ **4.**

(a) Give an example of a Galois extension $K/\mathbb{Q}$ with Galois group $\mathbb{Z}/4\mathbb{Z}$, and prove that it is such an example.

(b) Let $K = F_q(t)$, and let $L = F_q(t^{1/p})$ where q is a power of p. Prove that there are no automorphisms of L fixing K.

Solution 4.

(a) Consider $K = \mathbb{Q}(\zeta_5)$ where ζ_5 is a primitive 5th root of unity. Then ζ_5 satisfies $x^5 - 1 = (x-1)(x^4+x^3+x^2+x+1)$. Since $\zeta_5 \neq 1$, it must satisfy $f(x) = x^4+x^3+x^2+x+1$ (note that this is the cyclotomic polynomial $\phi_5(x)$). We will show that $f(x)$ is irreducible over $\mathbb{Q}$ to get that it is the minimal polynomial of ζ_5. Observe that $f(x+1) = x^4+5x^3+10x^2+10x+5$, which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion with $p = 5$. Thus, $f(x)$ is also irreducible. Then K is the splitting field of $f(x) = (x-\zeta_5)(x-\zeta_5^2)(x-\zeta_5^3)(x-\zeta_5^4)$, so $[K : \mathbb{Q}] = \deg f(x) = 4$, and $K/\mathbb{Q}$ is Galois since $f(x)$ is separable. Thus, $|\operatorname{Gal}(K/\mathbb{Q})| = 4$, and the automorphisms in $\operatorname{Gal}(K/\mathbb{Q})$ are given by $\zeta_5 \mapsto \zeta_5^i$ for $i = 1, 2, 3, 4$. Since each of the nontrivial automorphisms has order 4, we get that $\operatorname{Gal}(K/\mathbb{Q}) = \mathbb{Z}/4\mathbb{Z}$.

(b) Observe that $t^{1/p}$ satisfies $x^p - t \in K[x]$, and because K is characteristic p, this polynomial factors as $x^p - t = x^p - (t^{1/p})^p = (x - t^{1/p})^p$. Therefore, $t^{1/p}$ is its only root. Since L is the splitting field of $x^p - t$ over K, any automorphism of L over K is determined by how it permutes the roots of $x^p - t$. But since there is only one root, the only automorphism is the identity on L sending $t^{1/p} \mapsto t^{1/p}$.

∗ **5.**

Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in the variable x. Thus, x is transcendental over $\mathbb{C}$. Put

$$y = x^n + x^{-n} \in \mathbb{C}(x)$$

for some $n > 0$. Prove that the field extension $\mathbb{C}(x)/\mathbb{C}(y)$ is a finite Galois extension and find its degree.

Solution 5.

Let ζ be a primitive nth root of unity, let $\sigma_i : \mathbb{C}(x) \to \mathbb{C}(x)$ be the automorphism induced by $x \mapsto \zeta^i x$ for $0 \leq i \leq n-1$, and let τ be the automorphism defined by $x \mapsto \frac{1}{x}$. Notice that all of the σ_i and τ fix $\mathbb{C}(y)$ since $\zeta^n = 1$, so their products $\sigma_i\tau$ also fix $\mathbb{C}(y)$. Therefore, $Id, \sigma_1, \ldots, \sigma_{n-1}, \tau, \sigma_1\tau, \ldots, \sigma_{n-1}\tau$ are all elements of $\operatorname{Aut}(\mathbb{C}(x)/\mathbb{C}(y))$, so $|\operatorname{Aut}(\mathbb{C}(x)/\mathbb{C}(y))| \geq 2n$. On the other hand, we have that x is a root of $t^{2n} + 1 - (x^n + x^{-n})t^n \in \mathbb{C}(y)[t]$, which is a degree $2n$ polynomial, so $[\mathbb{C}(x) : \mathbb{C}(y)] \leq 2n$. Since $|\operatorname{Aut}(\mathbb{C}(x)/\mathbb{C}(y))| \leq [\mathbb{C}(x) : \mathbb{C}(y)]$, they must both be equal to $2n$. Thus, $\mathbb{C}(x)/\mathbb{C}(y)$ is a Galois extension of degree $2n$.

$*$ **6.**

(a) Let E be the splitting field over $\mathbb{Q}$ of $e(x) = x^3 + 2x^2 + 3x + 4$, which has discriminant -200. Find $\mathrm{Gal}(E/\mathbb{Q})$.

(b) Let F be the splitting field over $\mathbb{Q}$ of $f(x) = x^3 + 4x^2 + 7x + 6$, which has discriminant -72. Find $\mathrm{Gal}(F/\mathbb{Q})$.

(c) Find $\mathrm{Gal}(EF/\mathbb{Q})$.

Solution 6.

(a) First, check to see if $e(x)$ has any rational roots. The possible rational roots are $\pm 1, \pm 2, \pm 4$, none of work actually work. Since this is a degree 3 polynomial with no roots in $\mathbb{Q}$, this means that it is irreducible over $\mathbb{Q}$, so $\mathrm{Gal}(E/\mathbb{Q})$ is a transitive subgroup of S_3. So, it is either A_3 or S_3. Also, since $e'(x) = 3x^2 + 4x + 3$ has no real roots, $e(x)$ has no relative extrema, so $e(x)$ must just have one real root. This means that the other two roots are complex conjugates, so complex conjugation is an order 2 element of $\mathrm{Gal}(E/\mathbb{Q})$. Thus, $\mathrm{Gal}(E/\mathbb{Q}) = S_3$.

Another way to get that $\mathrm{Gal}(E/\mathbb{Q}) \neq A_3$ is because $\mathrm{Gal}(E/\mathbb{Q}) \subseteq A_3$ iff the discriminant of $e(x)$ is a square in $\mathbb{Q}$. Since -200 is not a square, we get that $\mathrm{Gal}(E/\mathbb{Q}) \not\subseteq A_3$.

(b) Using the rational root theorem again, we find that -2 is a root of $f(x)$, and it factors as $f(x) = (x+2)(x^2+2x+3)$. Therefore, the roots of $f(x)$ are $-2, -1+\sqrt{-2}, -1-\sqrt{-2}$. This gives that $F = \mathbb{Q}(\sqrt{-2})$, so $\mathrm{Gal}(F/\mathbb{Q}) = \mathbb{Z}/2\mathbb{Z}$.

(c) By definition, the discriminant of $e(x)$ is $[(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_3)]^2$ where $\alpha_1, \alpha_2, \alpha_3$ are the roots of $e(x)$. Therefore, the square root of the discriminant is an element in E. This gives that $\sqrt{-200} = 10\sqrt{-2} \in E$, so in particular, $F = \mathbb{Q}(\sqrt{-2}) \subseteq E$. Therefore, $EF = E$ and $\mathrm{Gal}(EF/\mathbb{Q}) = S_3$.

$*$ **7.**

Let K be the splitting field of the polynomial $f(x) = x^4 - x^2 - 1$ over $\mathbb{Q}$. Compute $\mathrm{Gal}(K/\mathbb{Q})$. (For partial credit, find the degree $[K : \mathbb{Q}]$.)

Solution 7.

Since $f(x)$ has degree 4, we know that $|\mathrm{Gal}(K/\mathbb{Q})| \leq 4!$ and that $\mathrm{Gal}(K/\mathbb{Q})$ is a transitive subgroup of S_4. So, it could be $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, D_4, A_4, or S_4. To find the roots of $f(x)$, notice that it is a quadratic in disguise. Letting $t = x^2$, we have that $f(t) = t^2 - t - 1$, which has roots $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Therefore, the roots of $f(x)$ are given by $x = \pm\sqrt{\alpha}, \pm\sqrt{\beta}$, and thus, $K = \mathbb{Q}(\sqrt{\alpha}, \sqrt{\beta})$.

Observe that K contains the quadratic subfield $\mathbb{Q}(\alpha) = \mathbb{Q}(\beta) = \mathbb{Q}(\sqrt{5})$, so the Galois group must contain an index 2 subgroup. Therefore, it cannot be A_4. Also, since $K = \mathbb{Q}(\sqrt{\alpha}, \sqrt{\beta})$ has degree at most 4 over $\mathbb{Q}(\sqrt{5})$, the Galois group of $K/\mathbb{Q}$ can have order at most 8. So, it cannot be S_4.

We will now show that $\mathrm{Gal}(K/\mathbb{Q}) = D_4$, a group of order 8, by showing that $[K : \mathbb{Q}] > 4$, which will eliminate $\mathbb{Z}/4\mathbb{Z}$ and the Klein 4 group as options. Consider the subfield $L = \mathbb{Q}(\sqrt{\alpha}, \beta) \subseteq K$, which is not all of K since $L \subseteq \mathbb{R}$ and $K \not\subseteq \mathbb{R}$. We will show that $[L : \mathbb{Q}] = 4$, which implies that $[K : \mathbb{Q}] > 4$. Since $\mathbb{Q}(\sqrt{5}) \subseteq L$, it is enough to show that L is strictly larger than $\mathbb{Q}(\sqrt{2})$. Suppose instead that $\sqrt{\alpha} = a + b\sqrt{5}$ with $a, b \in \mathbb{Q}$. Then squaring both sides gives

$$\frac{1}{2} + \frac{1}{2}\sqrt{5} = a^2 + 5b^2 + 2ab\sqrt{5}$$

which implies that $\frac{1}{2} = a^2 + 5b^2$ and $\frac{1}{2} = 2ab$. Substituting $b = \frac{1}{4a}$ into the first equation and simplifying (since $a \neq 0$) gives $16a^4 - 8a^2 + 5 = 0$. You can then use the rational root theorem to show that none of the possible roots in $\mathbb{Q}$ are actually roots. Therefore, $\sqrt{\alpha} \notin \mathbb{Q}(\sqrt{5})$, so $[L : \mathbb{Q}] = 4$, and we are done.